FermatQuinticJacobian: Lean 4 Certificate for the Jacobian Smoothness of the Fermat Quintic Fourfold

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/FermatQuinticJacobian
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Abstract

Standalone Lean 4 formalization (‘FermatQuinticJacobian.lean‘) certifying the non-degeneracy of the partial derivatives and Jacobian smoothness criterion for the Fermat quintic fourfold V(x_0^5 + ... + x_5^5) subset P^5.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Certifies Jacobian smoothness of the Fermat quintic host variety in Lean 4. Explicitly states that Jacobian smoothness is a prerequisite geometric property and not a resolution of the Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/FermatQuinticJacobian`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652994`
- **Earliest Git Commit SHA:** `777a2e97451ec4a366c03bbacb4333c001bb833d` (2026-10-03 12:49:57 PM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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A Lean 4 certificate, checked against Mathlib ‘v4.22.0‘, that the Fermat quintic
“‘text
G = X0^5 + X1^5 + X2^5 + X3^5 + X4^5 + X5^5
“‘
has partial derivatives
“‘text
pderiv i G = 5 * X i ^ 4
“‘
Those forms vanish together only at the origin of affine 6-space. By the Jacobian criterion, ‘V(G)‘ in ‘P^5‘ is smooth.
The theorem in the repository is ‘Hodge.FermatQuintic.partial_eq‘.
This is not a proof of the Hodge conjecture. It does not construct a cycle from a

Hodge class. It does not show that ‘H^{3,1}‘ vanishes: on this fourfold that primitive piece is the degree-4 part of ‘C[X_i]/(X_i^4)‘, of dimension 120.
The non-homogeneous expression
‘sum x_i^5 - 5 x0 x1 x2 x3 x4 x5‘ is not used. The product of six variables has

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/FermatQuinticJacobian` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFICATE.json`) and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
`FermatQuinticJacobian.lean`	1871 B	`d94c40a67456932d...`
`LICENSE`	11355 B	`9da72c2bee4f20ba...`
`NOTICE`	638 B	`1d6c88adb668f721...`
`README.md`	1106 B	`775cf9e9962b7aa2...`
`lakefile.toml`	288 B	`3f8f4cf04ccf5e35...`
`lean-toolchain`	25 B	`8f6680f8389e8ade...`

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